

PANDUAN FINAL MEMBUAT MINIGAME PLATFORMER SIDESCROLLER DI CONSTRUCT 2

Panduan ini disusun untuk membantu pembuatan minigame platformer sidescroller sederhana menggunakan **Construct 2**. Versi final ini sudah mencakup persiapan asset, setup object, behavior, keyboard input, touch input, animasi player, animasi musuh, animasi coin, animasi nyawa, sistem skor, sistem nyawa, sistem menyerang musuh dengan menginjak (stomp), game over, dan kondisi menang.

1. Tujuan Proyek

Minigame yang akan dibuat adalah game platformer 2D sidescroller sederhana dengan fitur berikut:

- Player dapat bergerak ke kiri dan ke kanan.
 - Player dapat melompat.
 - Player bisa dimainkan menggunakan keyboard maupun touch control.
 - Player dapat mengambil coin untuk menambah skor.
 - Player memiliki nyawa.
 - Player dapat **membunuh musuh dengan menginjak dari atas (stomp)**.
 - Player akan kehilangan nyawa jika menyentuh musuh dari samping atau bawah.
 - Musuh berjalan bolak-balik.
 - Terdapat animasi pada player, musuh, coin, dan heart.
 - Terdapat background music dan sound effect.
 - Game akan selesai saat semua coin habis diambil.
 - Game over terjadi saat nyawa habis.
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2. Asset Gratis yang Direkomendasikan

Berikut beberapa asset gratis yang dapat digunakan.

2.1 Karakter Player

Nama Asset	Link Download	Keterangan
Pixel Adventure (Pixel Frog)	https://pixelfrog-assets.itch.io/pixel-adventure-1	Karakter animasi lengkap: idle, run, jump, fall
FREE - 2D Pixel Art Male Character	https://gandalfhardcore.itch.io/free-2d-pixel-art-male-and-female-character-sidescroller	Karakter pria dan wanita, animasi run, jump, idle
Fantasy Knight	https://aamatniekss.itch.io/fantasy-knight-free-pixelart-animated-character	Knight animasi lengkap gratis
Warrior Free Animation	https://clembood.itch.io/warrior-free-animation-set	Animasi sidescroller warrior

2.2 Musuh

Nama Asset	Link Download	Keterangan
Kings and Pigs (Pixel Frog)	https://pixelfrog-assets.itch.io/kings-and-pigs	Musuh pig animasi lengkap
Monsters Creatures Fantasy	https://luizmelo.itch.io/monsters-creatures-fantasy	Berbagai jenis monster pixel art
Pixel Adventure 2	https://pixelfrog-assets.itch.io/pixel-adventure-2	Tambahan enemy sprite

2.3 Heart dan UI

Nama Asset	Link Download	Keterangan
Brackeys' Platformer Bundle	https://brackeysgames.itch.io/brackeys-platformer-bundle	UI dan heart icon
Free Game Assets GUI	https://free-game-assets.itch.io/free-2d-game-gui	Health bar, heart, coin counter
GameArt2D Freebies	https://www.gameart2d.com/freebies.html	GUI gratis termasuk heart dan icon

2.4 Background

Nama Asset	Link Download	Keterangan
Sunny Land	https://ansimuz.itch.io/sunny-land	Background layer dan tileset
Oak Woods	https://brullov.itch.io/oak-woods	Environment dengan parallax
Adventure Pack Grassland	https://anokolisa.itch.io/sidescroller-pixelart-sprites-gif-character	Background dan tileset grassland

2.5 Sound Effect dan Background Music

Nama Asset	Link Download	Keterangan
Mixkit	https://mixkit.co/free-sound-effects/game/	SFX jump, coin, hurt, win
Freesound.org	https://freesound.org	Banyak sound effect gratis
OpenGameArt Audio	https://opengameart.org/content/platformer-sounds	SFX platformer lengkap
Pixabay Music		

Nama Asset	Link Download	Keterangan
	https://pixabay.com/music/search/chiptune/	BGM chiptune royalty-free

3. Setup Awal Project Construct 2

3.1 Membuat Project Baru

1. Buka **Construct 2**.
2. Pilih **File** → **New** → **New empty project**.
3. Atur ukuran layout, misalnya 800 × 450.
4. Simpan project dengan nama game yang diinginkan.

3.2 Import Asset

1. Klik kanan pada panel project.
2. Pilih **Insert New Object** atau import sprite sesuai kebutuhan.
3. Import asset berikut:
4. Player
5. Enemy
6. Coin
7. Heart
8. Platform / tileset
9. Background
10. Tombol touch
11. Untuk sprite sheet, buka **Edit animations** lalu masukkan frame animasi.

4. Object yang Dibutuhkan

Buat object berikut di layout:

Nama Object	Type	Fungsi
Player	Sprite	Karakter utama
Enemy	Sprite	Musuh

Nama Object	Type	Fungsi
Coin	Sprite	Item yang dikumpulkan
Platform	Sprite / Tiled Background	Lantai dan platform
Heart1	Sprite	Nyawa 1
Heart2	Sprite	Nyawa 2
Heart3	Sprite	Nyawa 3
ScoreText	Text	Menampilkan skor
Background	Tiled Background / Sprite	Latar belakang
BtnLeft	Sprite	Tombol touch kiri
BtnRight	Sprite	Tombol touch kanan
BtnJump	Sprite	Tombol touch lompat
WallMarker	Sprite	Penanda batas patroli musuh
Keyboard	Input plugin	Membaca keyboard
Touch	Input plugin	Membaca layar sentuh
Audio	Audio plugin	Musik dan sound effect

5. Layer yang Disarankan

Gunakan minimal 3 layer:

Layer	Isi	Catatan
Background	Langit, dekorasi, parallax	Bisa diberi efek scroll
Game	Player, enemy, coin, platform	Layer utama gameplay
HUD	Heart, score, tombol touch	Set parallax = 0,0

Penting: Layer HUD harus memiliki **Parallax 0,0** agar UI tetap diam di layar.

6. Behavior yang Digunakan

6.1 Player

Tambahkan behavior berikut pada object `Player`: - Platform - Scroll To - Flash (opsional)

6.2 Enemy

Tambahkan behavior berikut pada object `Enemy`: - Platform atau Bullet - Flash (opsional)

6.3 Platform

Tambahkan behavior: - Solid

7. Pengaturan Behavior Player

Pada object `Player`, atur behavior Platform seperti berikut:

Parameter	Nilai
Max speed	200
Acceleration	600
Deceleration	600
Jump strength	500
Gravity	1200
Max fall speed	800
Default controls	No

Default controls diset ke **No** karena kontrol keyboard dan touch akan dibuat manual lewat event sheet.

8. Variabel Global

Tambahkan global variable berikut:

Nama Variabel	Tipe	Nilai Awal	Fungsi
score	Number	0	Menyimpan skor
lives	Number	3	Menyimpan jumlah nyawa
isInvincible	Number	0	0 = normal, 1 = kebal
invincibleTimer	Number	0	Timer kebal sementara
touchLeft	Number	0	Tombol kiri disentuh
touchRight	Number	0	Tombol kanan disentuh
touchJump	Number	0	Tombol lompat disentuh

Catatan penting: di Construct 2, gunakan **Number** untuk status seperti on/off. Jadi jangan pakai Boolean untuk `isInvincible`.

9. Setup Animasi Sprite

9.1 Animasi Player

Pada object `Player`, buat animasi berikut:

Nama Animasi	Loop	Speed
idle	Yes	5
run	Yes	10
jump	No	1

Nama Animasi	Loop	Speed
fall	No	1
hurt	No	8

9.2 Animasi Enemy

Pada object `Enemy`, buat animasi berikut:

Nama Animasi	Loop	Speed	Keterangan
walk	Yes	8	Animasi patroli
idle	Yes	4	Animasi diam
hurt	No	6	Animasi saat diinjak
death	No	8	Animasi mati setelah diinjak

9.3 Animasi Coin

Pada object `Coin`, buat animasi berikut:

Nama Animasi	Loop	Speed
spin	Yes	10

9.4 Animasi Heart

Untuk object `Heart1`, `Heart2`, dan `Heart3`, gunakan 2 frame:

- Frame 0 = penuh
- Frame 1 = kosong

Dengan cara ini, heart bisa berubah sesuai jumlah nyawa.

10. Event Sheet Logic Final

Di bawah ini adalah susunan event final yang disarankan.

10.1 Start Layout

```
EVENT 1:
  CONDITION: System → On start of layout
  ACTION 1: Audio → Play music "bgm_main" looping Yes
  ACTION 2: Set score = 0
  ACTION 3: Set lives = 3
  ACTION 4: Set isInvincible = 0
  ACTION 5: Set invincibleTimer = 0
  ACTION 6: Set touchLeft = 0
  ACTION 7: Set touchRight = 0
  ACTION 8: Set touchJump = 0
```

10.2 Keyboard Input

Karena default control dimatikan, gunakan event berikut:

```
EVENT 2:
  CONDITION: Keyboard → Key is down → Left Arrow
  ACTION:   Player → Platform → Simulate control Left

EVENT 3:
  CONDITION: Keyboard → Key is down → Right Arrow
  ACTION:   Player → Platform → Simulate control Right

EVENT 4:
  CONDITION: Keyboard → On key pressed → Space
  ACTION:   Player → Platform → Simulate control Jump
```

Tambahan tombol WASD:

```
EVENT 5:
  CONDITION: Keyboard → Key is down → A
  ACTION:   Player → Platform → Simulate control Left
```

EVENT 6:

CONDITION: Keyboard → Key is down → D

ACTION: Player → Platform → Simulate control Right

EVENT 7:

CONDITION: Keyboard → On key pressed → W

ACTION: Player → Platform → Simulate control Jump

10.3 Touch Input

Buat 3 tombol pada layer HUD: - BtnLeft - BtnRight - BtnJump

Atur posisi di bawah layar.

Reset status touch

EVENT 8:

CONDITION: System → Every tick

ACTION 1: Set touchLeft = 0

ACTION 2: Set touchRight = 0

ACTION 3: Set touchJump = 0

Cek sentuhan tombol

EVENT 9:

CONDITION: Touch → Is in touch with BtnLeft

ACTION: Set touchLeft = 1

EVENT 10:

CONDITION: Touch → Is in touch with BtnRight

ACTION: Set touchRight = 1

EVENT 11:

CONDITION: Touch → Is in touch with BtnJump

ACTION: Set touchJump = 1

Simulasikan touch ke player

EVENT 12:

CONDITION: touchLeft = 1

ACTION: Player → Platform → Simulate control Left

EVENT 13:

CONDITION: touchRight = 1

ACTION: Player → Platform → Simulate control Right

EVENT 14:

CONDITION: touchJump = 1

ACTION: Player → Platform → Simulate control Jump

10.4 Animasi Player

Urutan animasi harus benar. Event hurt diletakkan paling atas agar tidak tertimpa animasi lain.

EVENT 15:

CONDITION 1: isInvincible = 1

CONDITION 2: Player → Is on floor

ACTION: Player → Set animation "hurt"

EVENT 16:

CONDITION 1: Player → Platform → Is moving

CONDITION 2: Player → Platform → Is on floor

CONDITION 3: isInvincible = 0

ACTION: Player → Set animation "run"

EVENT 17:

CONDITION 1: Player → Platform → Is on floor

CONDITION 2: Player → Platform → Vector X = 0

```
CONDITION 3: isInvincible = 0  
ACTION:      Player → Set animation "idle"
```

```
EVENT 18:  
CONDITION 1: Player → Platform → Is jumping  
CONDITION 2: isInvincible = 0  
ACTION:      Player → Set animation "jump"
```

```
EVENT 19:  
CONDITION 1: Player → Platform → Is falling  
CONDITION 2: isInvincible = 0  
ACTION:      Player → Set animation "fall"
```

Arah hadap player

```
EVENT 20:  
CONDITION: Keyboard → Key is down → Left Arrow  
ACTION:    Player → Set mirrored  
  
EVENT 21:  
CONDITION: Keyboard → Key is down → Right Arrow  
ACTION:    Player → Set not mirrored  
  
EVENT 22:  
CONDITION: touchLeft = 1  
ACTION:    Player → Set mirrored  
  
EVENT 23:  
CONDITION: touchRight = 1  
ACTION:    Player → Set not mirrored
```

10.5 Gerak dan Animasi Musuh

Tambahkan instance variable pada enemy:

Nama	Tipe	Nilai Awal
direction	Number	1
isDead	Number	0

Patrol musuh

EVENT 24:

CONDITION 1: Enemy.direction = 1

CONDITION 2: Enemy.isDead = 0

ACTION 1: Enemy → Platform → Simulate control Right

ACTION 2: Enemy → Set not mirrored

ACTION 3: Enemy → Set animation "walk"

EVENT 25:

CONDITION 1: Enemy.direction = -1

CONDITION 2: Enemy.isDead = 0

ACTION 1: Enemy → Platform → Simulate control Left

ACTION 2: Enemy → Set mirrored

ACTION 3: Enemy → Set animation "walk"

EVENT 26:

CONDITION: Enemy → On collision with WallMarker

ACTION: Enemy → Set direction = direction * -1

10.6 Menyerang Musuh dengan Menginjak (Stomp)

Sistem stomp mendeteksi apakah player **jatuh dari atas** dan posisi bawah player lebih tinggi dari tengah musuh. Jika ya, musuh mati. Jika tidak, player yang kehilangan nyawa.

Penjelasan logika stomp:

Stomp terdeteksi jika **dua kondisi terpenuhi secara bersamaan**: 1. Player sedang dalam kondisi **falling** (jatuh ke bawah / Vector Y > 0) 2. Posisi bawah player (Player.BBoxBottom) berada di **atas tengah musuh** (Enemy.Y)

Jika kondisi ini tidak terpenuhi, artinya player menyentuh musuh dari samping atau bawah → player kehilangan nyawa.

Instance variable tambahan yang dibutuhkan pada Enemy:

- isDead : Number, nilai awal = 0

Event Stomp — Membunuh Musuh:

```
EVENT STOMP 1:
  CONDITION 1: Player → On collision with Enemy
  CONDITION 2: Player → Platform → Vector Y > 0      (player
sedang jatuh)
  CONDITION 3: Player.BBoxBottom < Enemy.Y + 20        (kaki player
di atas tengah musuh)
  CONDITION 4: Enemy.isDead = 0                        (musuh belum
mati)

  ACTION 1:  Enemy → Set isDead = 1
  ACTION 2:  Enemy → Set animation "death" (from beginning: Yes)
  ACTION 3:  Enemy → Platform → Set enabled → No      (hentikan
gerakan musuh)
  ACTION 4:  Player → Platform → Set vector Y → -400  (player
memantul ke atas)
  ACTION 5:  Add 20 to score                            (bonus skor
stomp)
  ACTION 6:  Audio → Play "sfx_stomp"

--- SUB-EVENT ---
CONDITION: System → Wait for animation "death" to finish
```

```
(atau pakai timer manual di bawah)
ACTION:      Enemy → Destroy
```

💡 **Alternatif** jika Construct 2 tidak support wait animation: gunakan timer manual.

Alternatif menggunakan timer:

Tambahkan instance variable pada Enemy: - deathTimer : Number, nilai awal = 0

```
EVENT STOMP 2 (timer death):
  CONDITION: Enemy.isDead = 1
  ACTION:    Add dt to Enemy.deathTimer

  --- SUB-EVENT ---
  CONDITION: Enemy.deathTimer >= 0.6      (durasi animasi death
~0.6 detik)
  ACTION:    Enemy → Destroy
```

Event Kena Musuh dari Samping/Bawah (Player Kalah):

```
EVENT STOMP 3:
  CONDITION 1: Player → On collision with Enemy
  CONDITION 2: isInvincible = 0
  CONDITION 3: Enemy.isDead = 0

  // Pastikan ini BUKAN stomp (player tidak sedang jatuh atau
posisi tidak tepat)
  CONDITION 4: Player → Platform → Vector Y <= 0
                OR Player.BBoxBottom >= Enemy.Y + 20

  ACTION 1: Subtract 1 from lives
  ACTION 2: Set isInvincible = 1
  ACTION 3: Set invincibleTimer = 0
```

```
ACTION 4: Audio → Play "sfx_hurt"
```

```
ACTION 5: Player → Flash
```

⚠ **Catatan urutan event:** Letakkan **EVENT STOMP 1** di atas **EVENT STOMP 3** di event sheet agar stomp diperiksa lebih dulu sebelum logika kena musuh dijalankan.

Cara Alternatif yang Lebih Sederhana (Tanpa Kondisi Kompleks):

Jika ingin cara lebih mudah, bisa pisahkan dengan **2 collision box berbeda** pada musuh:

1. Buat collision polygon musuh bagian **atas** (kepala) → trigger stomp
2. Buat collision polygon musuh bagian **samping/bawah** (badan) → trigger hurt player

Atau gunakan pendekatan overlap:

```
EVENT STOMP (simple):
```

```
CONDITION 1: Player → Is overlapping Enemy
```

```
CONDITION 2: Player → Platform → Is falling
```

```
CONDITION 3: Player.Y + Player.Height/2 < Enemy.Y
```

```
CONDITION 4: Enemy.isDead = 0
```

```
ACTION 1: Enemy → Set isDead = 1
```

```
ACTION 2: Enemy → Set animation "death" (from beginning: Yes)
```

```
ACTION 3: Player → Platform → Set vector Y → -400
```

```
ACTION 4: Add 20 to score
```

```
ACTION 5: Audio → Play "sfx_stomp"
```

10.7 Coin dan Skor

Coin akan beranimasi otomatis jika animasi `spin` di-set loop.

```
EVENT 27:
```

```
CONDITION: Player → On collision with Coin
```

```
ACTION 1: Add 10 to score
```



```
ACTION 2: Audio → Play "sfx_coin"
ACTION 3: Coin → Destroy
```

10.8 Kena Musuh dan Invincibility

```
EVENT 28:
    CONDITION 1: Player → On collision with Enemy
    CONDITION 2: isInvincible = 0
    ACTION 1: Subtract 1 from lives
    ACTION 2: Set isInvincible = 1
    ACTION 3: Set invincibleTimer = 0
    ACTION 4: Audio → Play "sfx_hurt"
    ACTION 5: Player → Flash
```

```
EVENT 29:
    CONDITION: isInvincible = 1
    ACTION: Add dt to invincibleTimer
```

```
EVENT 30:
    CONDITION: invincibleTimer >= 2
    ACTION 1: Set isInvincible = 0
    ACTION 2: Set invincibleTimer = 0
```

10.9 Update Nyawa Heart

Masing-masing heart mempunyai 2 frame: - frame 0 = penuh - frame 1 = kosong

```
EVENT 31:
    CONDITION: lives >= 1
    ACTION: Heart1 → Set frame 0
    ELSE: Heart1 → Set frame 1

EVENT 32:
```

```
CONDITION: lives >= 2
ACTION:    Heart2 → Set frame 0
ELSE:      Heart2 → Set frame 1
```

EVENT 33:

```
CONDITION: lives >= 3
ACTION:    Heart3 → Set frame 0
ELSE:      Heart3 → Set frame 1
```

10.10 Update Score Text

EVENT 34:

```
CONDITION: System → Every tick
ACTION:    ScoreText → Set text to "Score: " & score
```

10.11 Game Over

EVENT 35:

```
CONDITION: lives <= 0
ACTION 1:  Audio → Play "sfx_gameover"
ACTION 2:  Wait 1 second
ACTION 3:  Go to layout "GameOver"
```

10.12 Menang / Level Clear

EVENT 36:

```
CONDITION: Coin → Count = 0
ACTION 1:  Audio → Play "sfx_win"
ACTION 2:  Wait 1 second
ACTION 3:  Go to layout "WinScreen"
```

11. Urutan Event yang Direkomendasikan

Susun event di event sheet dengan urutan berikut agar tidak bentrok:

1. On start of layout
 2. Reset touch variable
 3. Cek sentuhan tombol
 4. Keyboard input
 5. Simulate touch control
 6. **Collision stomp (bunuh musuh dari atas) ← Letakkan SEBELUM collision musuh biasa**
 7. Collision enemy (player kena musuh dari samping/bawah)
 8. Collision coin
 9. Timer invincible
 10. Animasi player
 11. Animasi enemy (patroli)
 12. Timer kematian musuh (deathTimer)
 13. Update heart
 14. Update score
 15. Game over
 16. Win condition
-

12. Struktur Layout yang Disarankan

```
Project
├── Layout: MainMenu
│   └── Event Sheet: menuEvents
├── Layout: Level1
│   └── Event Sheet: gameEvents
├── Layout: Level2
│   └── Event Sheet: gameEvents
├── Layout: GameOver
│   └── Event Sheet: gameoverEvents
└── Layout: WinScreen
    └── Event Sheet: winEvents
```

Jika ada lebih dari satu level, gunakan **Event Sheet Include** supaya logic utama tidak perlu ditulis ulang.

13. Format Audio untuk Construct 2

Construct 2 biasanya memerlukan audio dalam format: - .ogg - .m4a

Untuk convert file audio, kamu bisa memakai: - Audacity - Convertio

Pisahkan file berikut: - SFX pendek ke folder **Sounds** - Musik panjang ke folder **Music**

Sound effect yang dibutuhkan untuk sistem stomp: - `sfx_stomp` — suara injak/lompatan mengenai musuh

14. Tips Implementasi

- Gunakan ukuran tombol touch yang cukup besar.
 - Gunakan opacity tombol touch sekitar 50% sampai 70%.
 - Tempatkan tombol kiri dan kanan di kiri bawah.
 - Tempatkan tombol lompat di kanan bawah.
 - Gunakan sprite yang konsisten gayanya, misalnya sesama pixel art.
 - Gunakan animasi run untuk enemy agar terlihat hidup.
 - Untuk sistem stomp, pastikan nilai toleransi posisi (`Enemy.Y + 20`) disesuaikan dengan ukuran sprite musuh.
 - Pantul player setelah stomp (`Set vector Y → -400`) membuat gameplay terasa responsif seperti game Mario.
 - Jangan terlalu banyak event yang sama agar project tetap rapi.
 - Simpan backup project secara berkala.
-

15. Checklist Sebelum Testing

- [] Player sudah punya behavior Platform dan Scroll To.
- [] Platform sudah punya behavior Solid.
- [] Enemy sudah punya event patroli.
- [] Enemy sudah punya instance variable `isDead` dan `deathTimer`.

- [] Semua global variable sudah dibuat.
 - [] isInvincible memakai Number, bukan Boolean.
 - [] Keyboard input sudah bekerja.
 - [] Touch input sudah bekerja.
 - [] Animasi player idle, run, jump, fall, hurt sudah ada.
 - [] Animasi enemy walk, idle, death sudah ada.
 - [] Animasi coin spin sudah ada.
 - [] Heart punya frame penuh dan kosong.
 - [] Score text muncul.
 - [] BGM dan SFX sudah masuk (termasuk sfx_stomp).
 - [] **Event stomp sudah diletakkan SEBELUM event collision musuh biasa.**
 - [] Musuh menampilkan animasi death lalu destroy setelah diinjak.
 - [] Player memantul ke atas setelah berhasil stomp.
 - [] Player kehilangan nyawa jika menyentuh musuh dari samping/bawah.
 - [] Event game over dan win sudah berjalan.
 - [] Layout HUD sudah memakai parallax 0,0.
-

16. Penutup

Dengan mengikuti panduan ini, kamu sudah bisa membuat minigame platformer sidescroller sederhana di Construct 2 yang dapat dimainkan di desktop maupun perangkat layar sentuh, lengkap dengan sistem **stomp** untuk membunuh musuh. Untuk pengembangan lanjutan, kamu bisa menambahkan level baru, checkpoint, double jump, serangan jarak jauh, particle effect saat stomp, atau menu pause.